

Assignment 2

Up/Down Counter — Verilog Design, Simulation

Module	Up_Dn_Counter
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Course	Digital Design

Specification Summary

- 5-bit up/down counter (Counter[4:0])
- Load has highest priority: on posedge CLK, if Load is high, Counter is loaded from IN
- Down has next priority: if not loading and Down is high, Counter decrements by 1
- Up has lowest priority: if not loading/decrementing and Up is high, Counter increments by 1
- Increment saturates at 31 (does not wrap); Decrement saturates at 0 (does not wrap)
- High flag: active high when Counter == 31
- Low flag: active high when Counter == 0

1. Verilog Design Code (Up_Dn_Counter.v)

```
module Up_Dn_Counter(
    input [4:0] IN,
    input Load,Up,Down,CLK,
    output reg [4:0] Counter,
    output High,Low
);
    assign High = (Counter == 5'd31);
    assign Low = (Counter == 5'd0);
    always @(posedge CLK) begin
        if(Load) begin
            Counter <= IN;
        end
        else if (Down && !Low) begin
            Counter <= Counter - 1;
        end
        else if (Up && !High) begin
            Counter <= Counter + 1;
        end
        else Counter <= Counter;
    end
endmodule
```

2. Verilog Testbench (Up_Dn_Counter_TB.v)

```
module Up_Dn_Counter_TB;
    reg [4:0] IN_TB;
    reg Load_TB,Up_TB,Down_TB,CLK_TB;
    wire [4:0] Counter_TB;
    wire High_TB,Low_TB;
    Up_Dn_Counter DUT_Up_Dn_Counter (
        .IN(IN_TB),
        .Load(Load_TB),
        .Up(Up_TB),
        .Down(Down_TB),
        .CLK(CLK_TB),
        .Counter(Counter_TB),
        .High(High_TB),
        .Low(Low_TB)
    );
    initial begin
        CLK_TB = 0;
        forever begin
            #5 CLK_TB = ~CLK_TB;
        end
    end
    initial begin
        $dumpfile("Up_Dn_Counter_TB.vcd");
        $dumpvars;
    end
    initial begin
        $monitor("At Time: %0t,Counter: %0b,High Flag: %0b,Low Flag: %0b",
            $time,Counter_TB,High_TB,Low_TB);
    end
    initial fork
        #0 {Load_TB,IN_TB} = 6'b110101;
        #10 {Load_TB,Down_TB,Up_TB} = 3'b111;
        #20 {Load_TB,Down_TB,Up_TB} = 3'b010;
        #30 {Load_TB,Down_TB,Up_TB} = 3'b011;
        #40 {Load_TB,Down_TB,Up_TB} = 3'b001;
        #50 {Load_TB,Down_TB,Up_TB} = 3'b000;
        #60 {Load_TB,IN_TB} = 6'b100001;
        #70 {Load_TB,Down_TB,Up_TB} = 3'b010;
        #110 {Load_TB,IN_TB} = 6'b111110;
        #130 {Load_TB,Down_TB,Up_TB} = 3'b001;
        #200 $finish;
    join
endmodule
```

3. Synthesis Diagram

The diagram below shows the synthesized structure of the design: a 5-bit register (D flip-flops) clocked by CLK, fed by a 3-level multiplexer chain that selects between the loaded value (IN), the incremented value (Counter+1, gated by Up and not High), the decremented value (Counter-1, gated by Down and not Low), and the held value (Counter). Load selects the top-priority mux; the +1/-1 paths are gated by AND gates with the High/Low flags to implement the saturation behavior. Two comparators ($= 31$ and $= 0$) generate the High and Low flags from the registered Counter output.

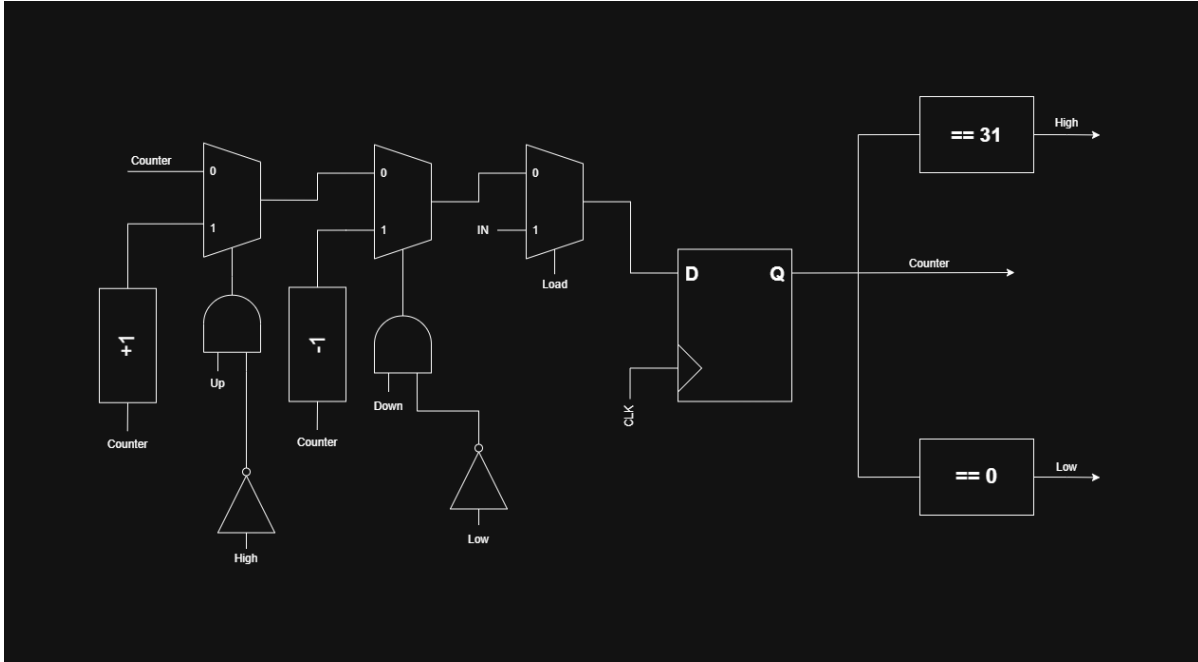


Figure 1: Synthesis (structural) diagram of Up_Dn_Counter, drawn in Draw.io

4. Test Case Summary

The testbench applies a free-running clock (period 10 time units) and drives the following stimulus sequence to Load, Down, Up, and IN. Results were obtained from an actual Icarus Verilog (iverilog/vvp) simulation run of the two files above.

Time	Load	Down	Up	IN	Purpose
0	1	-	-	10101	Load Counter with 21 (10101)
10	1	1	1	-	Load still asserted (highest priority) — Down/Up ignored
20	0	1	0	-	Down only: decrement (Down > Up priority)
30	0	1	1	-	Down asserted with Up: Down wins, decrement
40	0	0	1	-	Up only: increment
50	0	0	0	-	Hold: Counter unchanged
60	1	-	-	00001	Load Counter with 1 (00001)
70	0	1	0	-	Down only: decrement toward 0
110	1	-	-	11110	Load Counter with 30 (11110)
130	0	0	1	-	Up only: increment toward 31 saturation

5. Simulation Waveform

The waveform below (captured from ModelSim) shows CLK, Load, Down, Up, IN, the registered Counter output, and the High/Low flags across the full test sequence. Counter follows the load and increment/decrement behavior exactly as specified, and High/Low pulse correctly at the 31 and 0 boundary values.

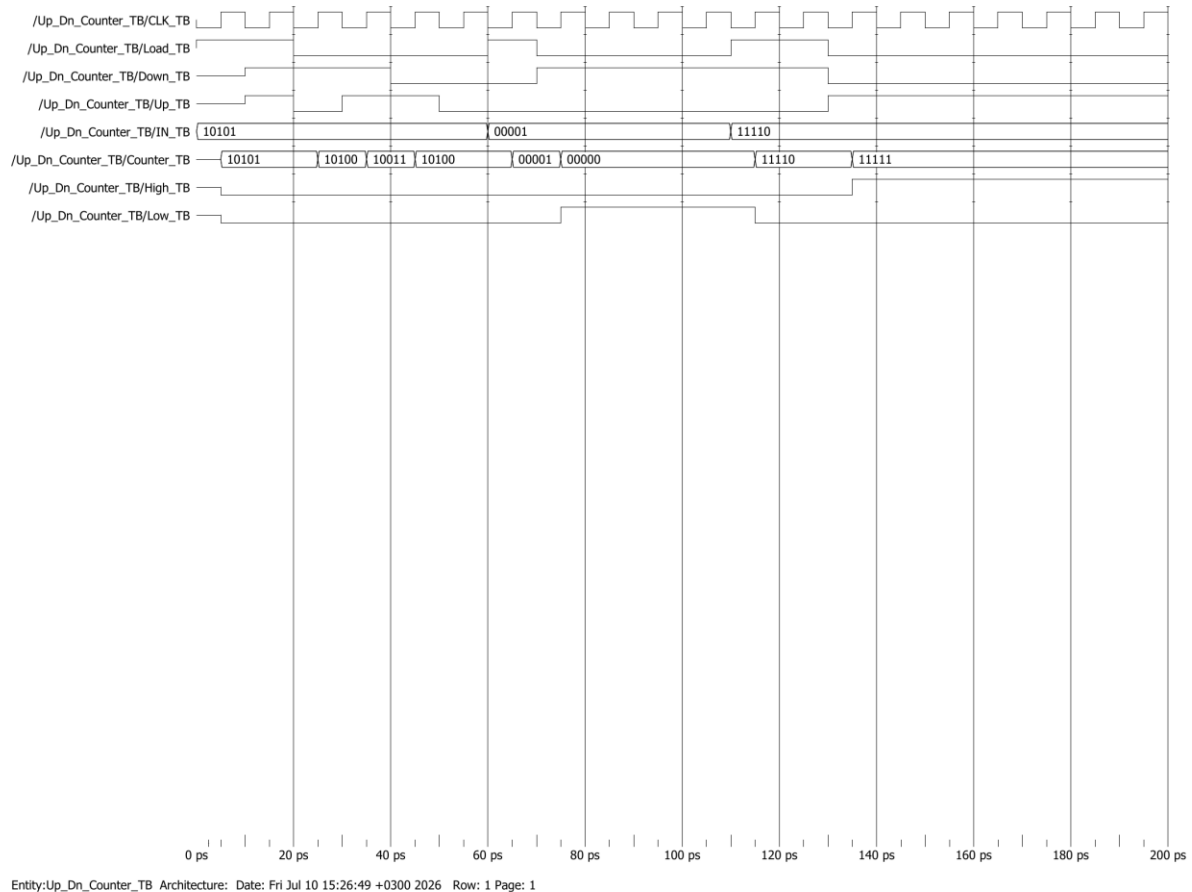


Figure 2: ModelSim simulation waveform for Up_Dn_Counter_TB

6. Simulation Transcript (Log)

Console transcript produced by running the testbench with Icarus Verilog (iverilog + vvp), driven by the \$monitor statement in the testbench:

```
VCD info: dumpfile Up_Dn_Counter_TB.vcd opened for output.
At Time: 0,Counter: xxxxx,High Flag: x,Low Flag: 1
At Time: 5,Counter: 10101,High Flag: 0,Low Flag: 1
At Time: 20,Counter: 10101,High Flag: 0,Low Flag: 0
At Time: 25,Counter: 10100,High Flag: 0,Low Flag: 0
At Time: 35,Counter: 10011,High Flag: 0,Low Flag: 0
At Time: 45,Counter: 10100,High Flag: 0,Low Flag: 0
At Time: 60,Counter: 10100,High Flag: 0,Low Flag: 1
At Time: 65,Counter: 1,High Flag: 0,Low Flag: 1
At Time: 70,Counter: 1,High Flag: 0,Low Flag: 0
At Time: 75,Counter: 0,High Flag: 0,Low Flag: 0
At Time: 110,Counter: 0,High Flag: 0,Low Flag: 1
At Time: 115,Counter: 11110,High Flag: 0,Low Flag: 1
At Time: 130,Counter: 11110,High Flag: 0,Low Flag: 0
At Time: 135,Counter: 11111,High Flag: 1,Low Flag: 0
Up_Dn_Counter_TB.v:40: $finish called at 200 (1s)
```

The transcript confirms: the counter is correctly loaded at t=5 (21), t=65 (1), and t=115 (30); decrements to 20, 19, 20 during t=25-45; correctly holds during t=55; decrements to 0 at t=75 with Low Flag asserted; and increments to 31 at t=135 with High Flag asserted, matching the specification.